COMPASS_TEMPEST_SGW_2022

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Sys.date

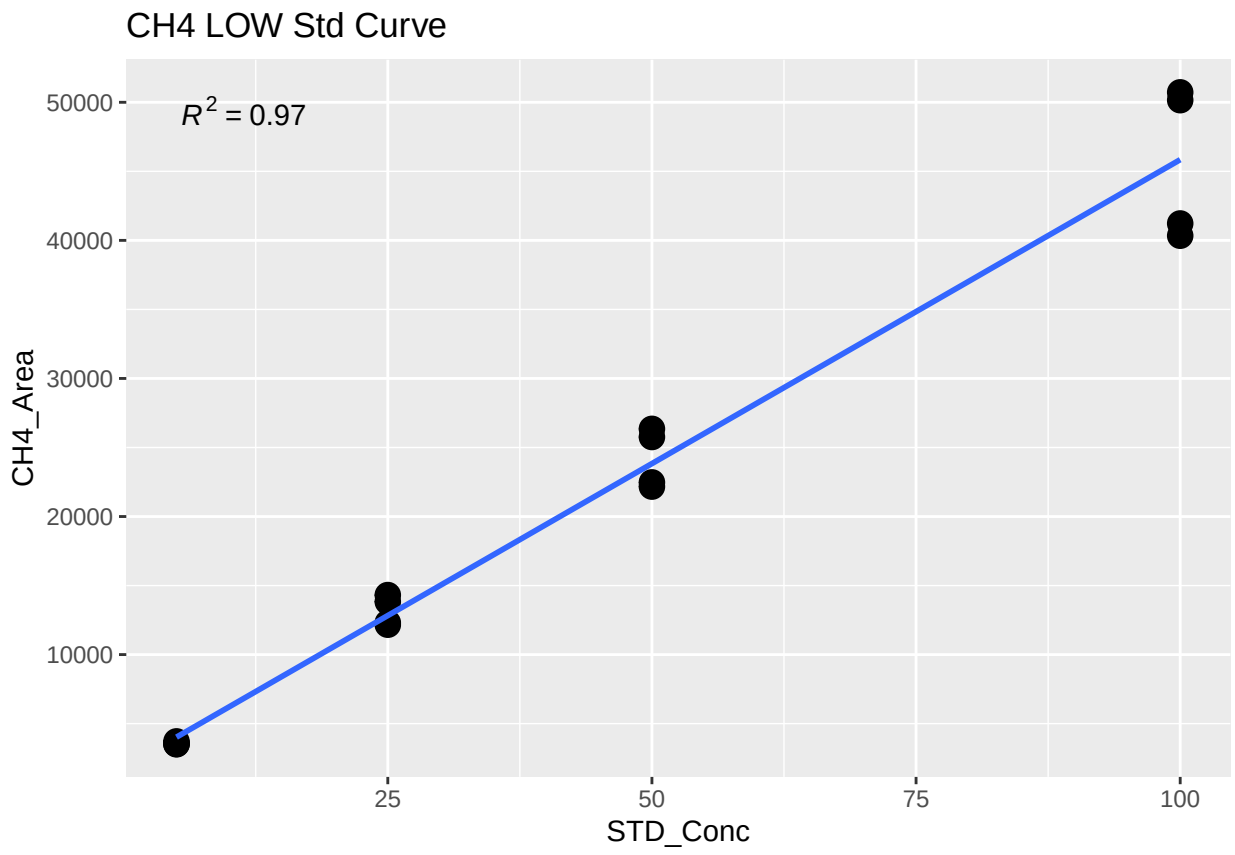
##Set Up

Read in first data file

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(c(STD_Conc, CO2_Area, CH4_Area), as.numeric)`.
## Caused by warning:
## ! NAs introduced by coercion
```

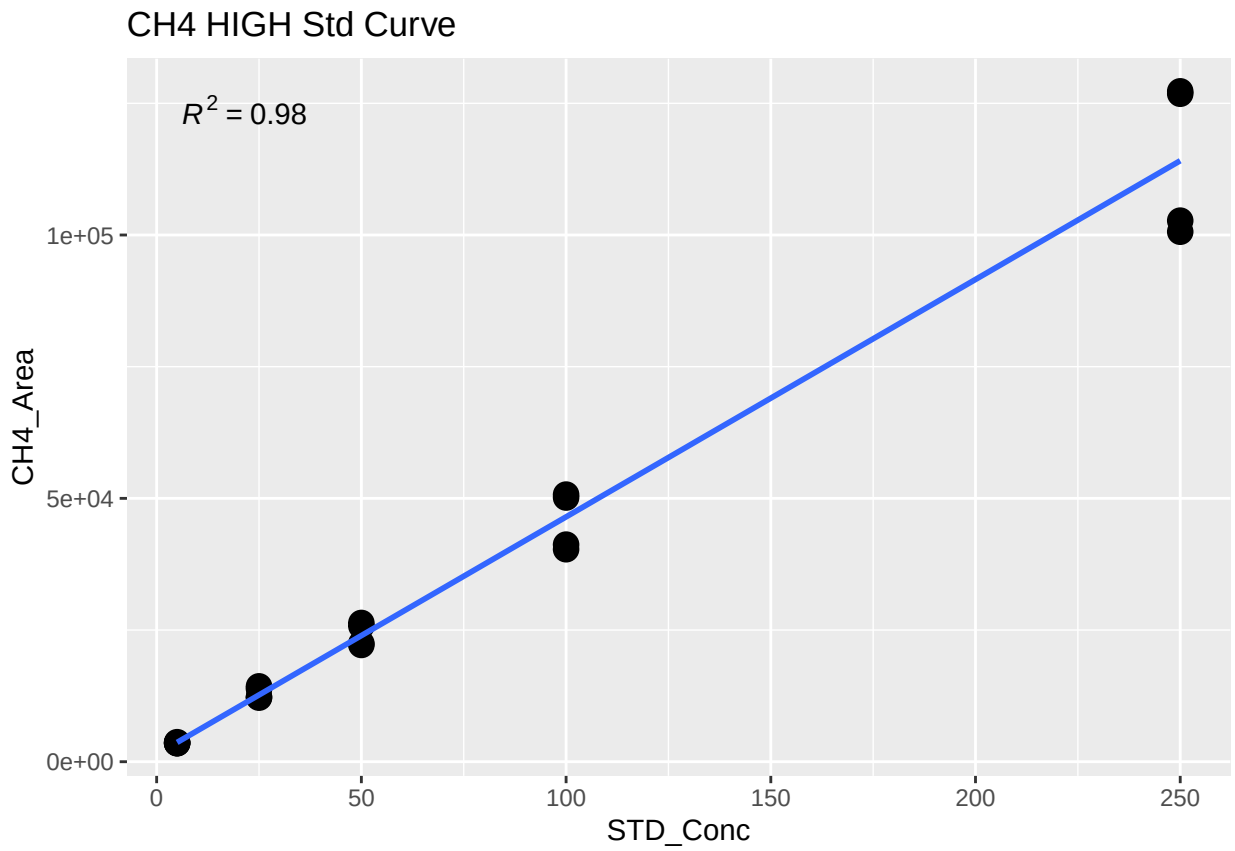
Assess standard curves

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5507.6  -837.4  -472.4  1594.6  4876.4
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              1820.70    1148.33   1.586    0.135
## stds_ch4_low$STD_Conc    440.24      20.03  21.981 2.98e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2847 on 14 degrees of freedom
## Multiple R-squared:  0.9718, Adjusted R-squared:  0.9698
## F-statistic: 483.2 on 1 and 14 DF,  p-value: 2.978e-12

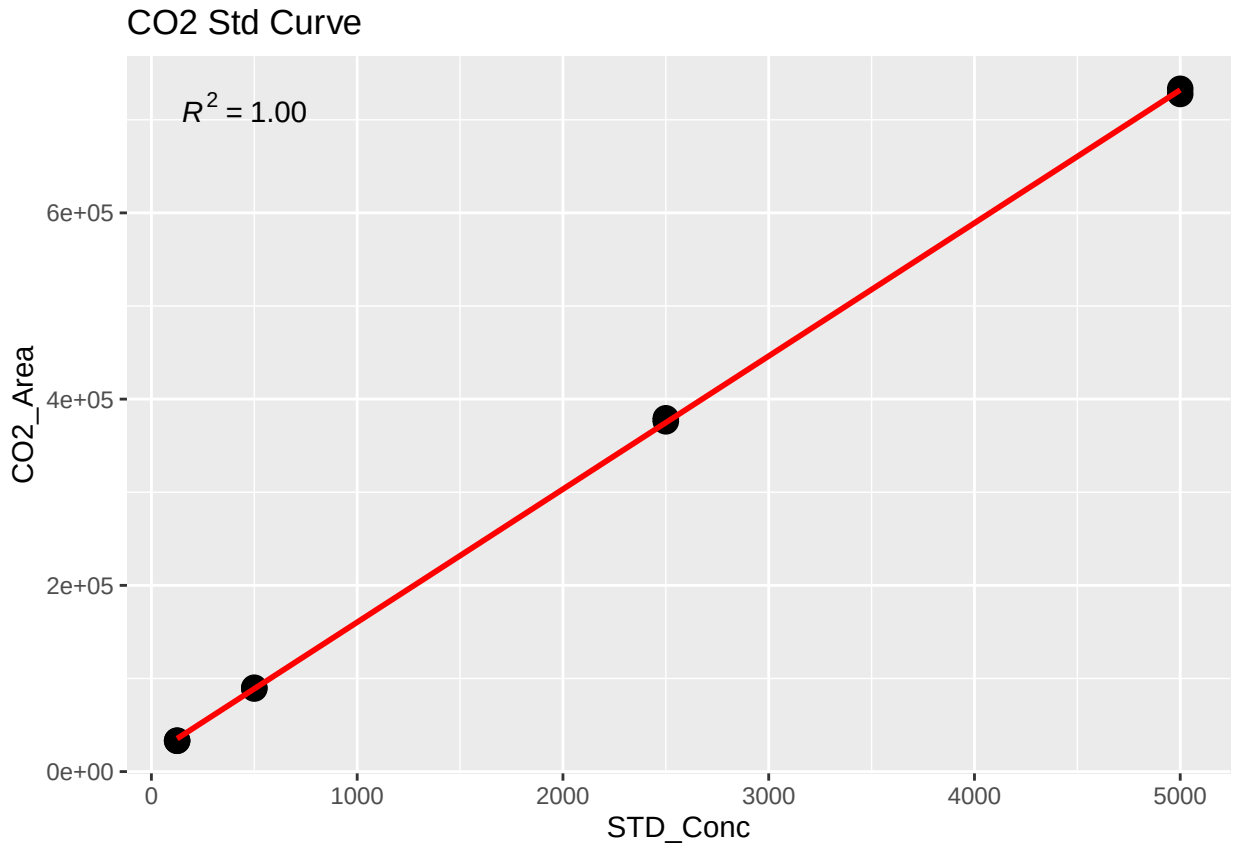
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13481.0  -1555.3   -116.2   1955.4  13179.0
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1411.86    2035.17   0.694    0.497
## stds_ch4$STD_Conc   450.76      16.55  27.244 4.38e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6507 on 18 degrees of freedom
## Multiple R-squared:  0.9763, Adjusted R-squared:  0.975
## F-statistic: 742.2 on 1 and 18 DF,  p-value: 4.385e-16

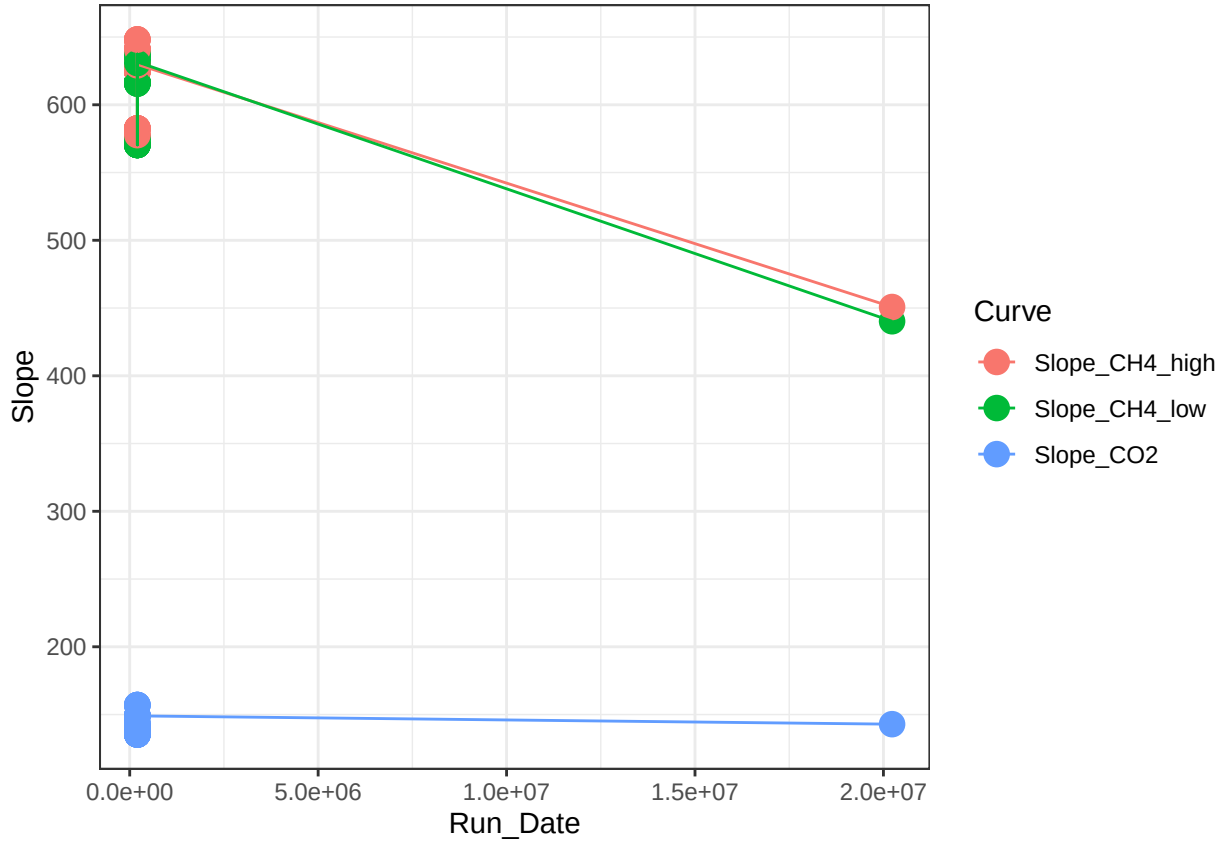
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4322.2 -2185.6   693.1  1287.3  4683.9
##
```

```
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.751e+04  1.555e+03   11.26 2.93e-05 ***
## stds_co2$STD_Conc 1.429e+02  5.538e-01  258.03 2.29e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3035 on 6 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 6.658e+04 on 1 and 6 DF,  p-value: 2.287e-13
```

```
##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 2 2 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 3 3      Slope_CO2 0.9999951 138.7384 5556.86838   202304
## 4 4  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 5 5 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 6 6      Slope_CO2 0.9999951 138.7384 5556.86838   202304
```



Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)
```

```r
#pull out samples
Samples <-  raw %>%
  filter(
    is.na(Sample_Type) |
    !str_detect(Sample_Type, "STD_CH4|STD_CO2|Blank|Chk_STD|CHKSTD|Check")
  )
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area >50721, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area >50721, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Slo

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)

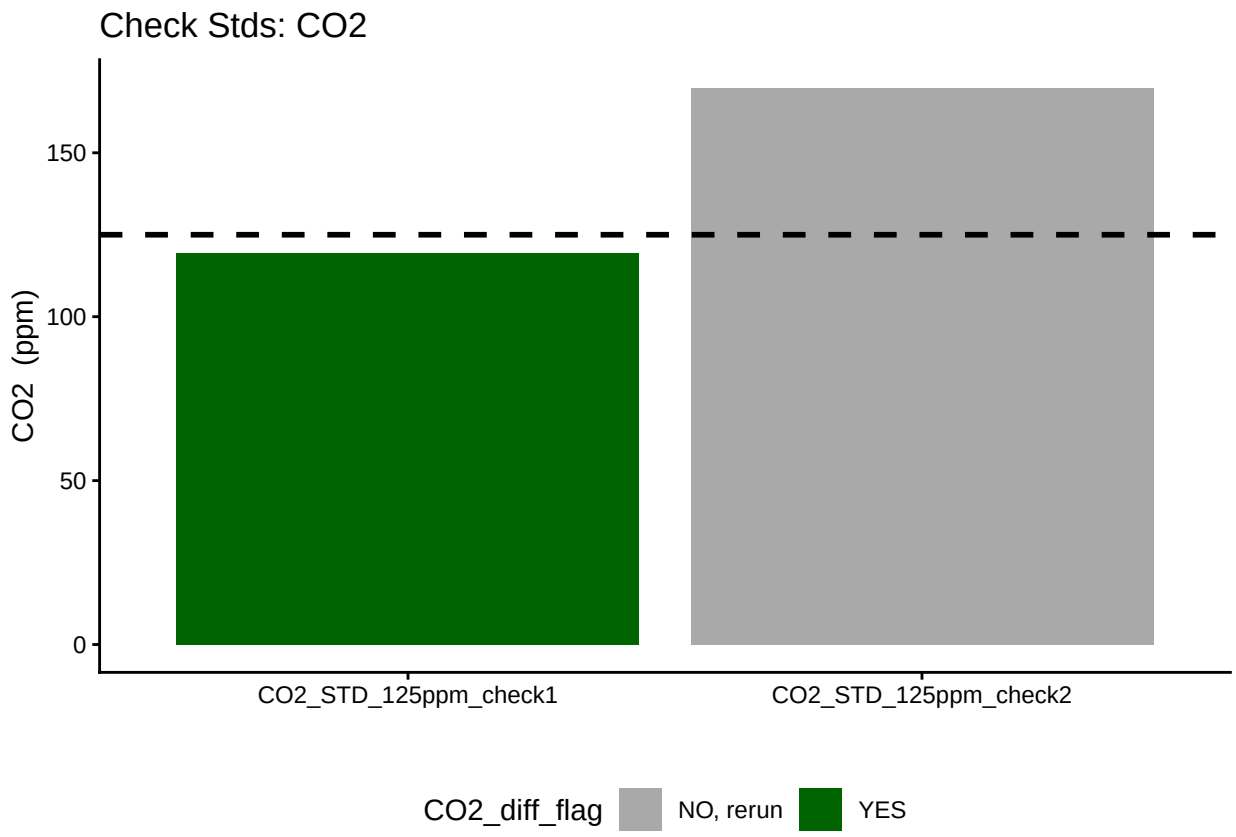
#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >250, "Needs Dilution", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >5000, "Needs Dilution", "Within Range")
#head(Samples)
```

Check the Check Standards

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Dilution correct samples

```
# #multiply the concentration by the dilution factor
# Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)
#
# Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

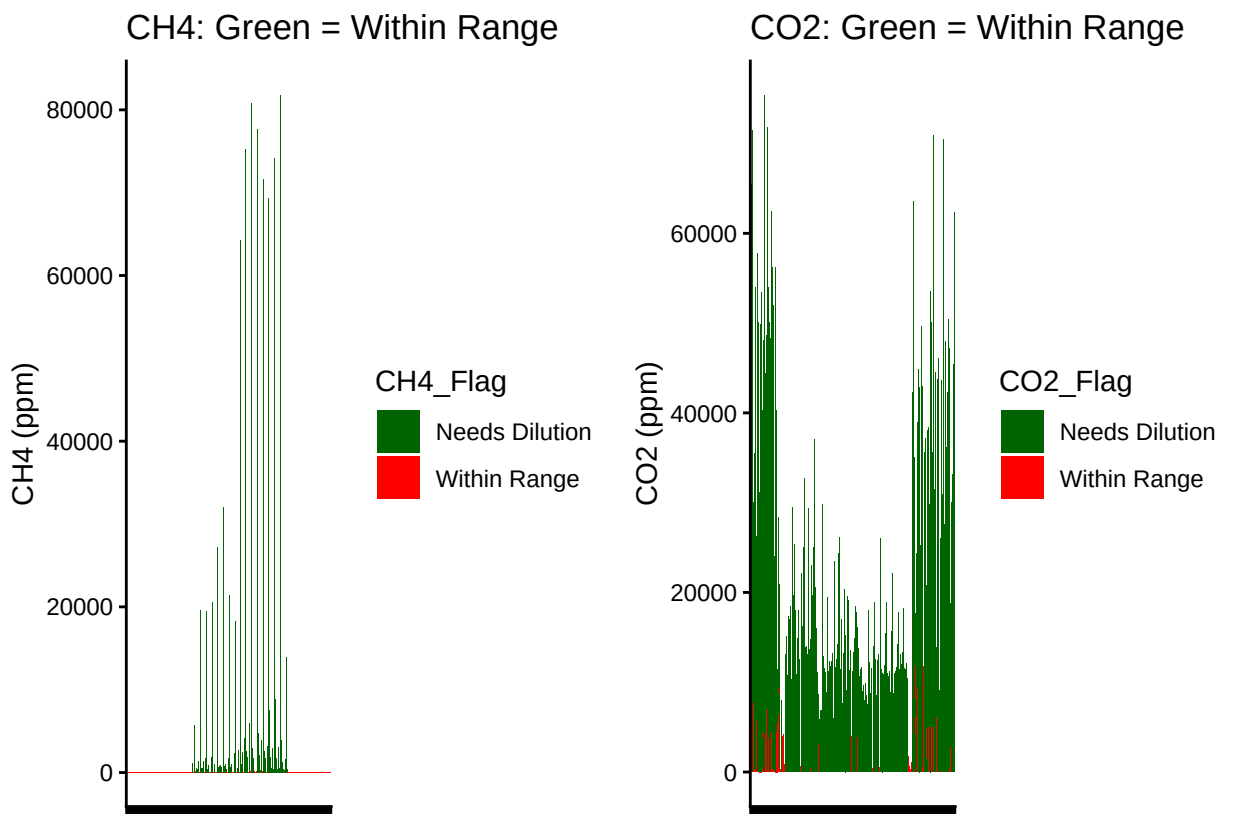
#check results
#head(Samples)

#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = `Sample ID`, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        theme_classic() +
        labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(axis.text.x = element_blank())
        # theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = `Sample ID`, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c( "darkgreen", "red"))+
        theme_classic() +
```

```
        labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
        theme(axis.text.x = element_blank())
        # theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```




```
##Percentage of samples that need dilution
Samples %>%
  summarise('Percent CO2 Samples Needing Dilution' = mean(CO2_Flag == "Needs Dilution", na.rm = TRUE) *
```

```
## # A tibble: 1 x 1
##   `Percent CO2 Samples Needing Dilution`
##                                    <dbl>
## 1                                   83.3
```

```
Samples%>%
  summarise('Percent CH4 Samples Needing Dilution' = mean(CH4_Flag == "Needs Dilution", na.rm = TRUE) *
```

```
## # A tibble: 1 x 1
##   `Percent CH4 Samples Needing Dilution`
##                                    <dbl>
## 1                                   16.2
```

If samples are water calculate gas in water

“‘{#r, echo=FALSE}

Convert ppm CH4 to total umol CH4 in the sample headspace

use Ideal Gas Law:

hdsp umol CH4 = ppm CH4*(PV/RT)

P = 1 (pressure in atm)

R = 0.082 (gas constant)

V = 0.015 (gas syringe volume in L)

T = “TempC” column value in Kelvin (equilibration temperature)

#set the volume used for the sample volume = 0.02 TempC <- 25 #I am assuming 25C not sure how we want to handle this

#Calculate CH4 concentrations in ppm Samples$CH4_Conc_umol <- ifelse(Samples$G_W == “W”, ((Samples$CH4_Conc_ppm_dilcorr*((1*volume/1000)/(0.08206*(TempC+273.15))))/(volume/1000)), ((((Samples$CH4_Conc_)) Samples$CO2_Conc_umol <- ifelse(Samples$G_W == “W”, ((Samples$CO2_Conc_ppm_dilcorr * ((1 * volume/1000)/(0.08206*(TempC+273.15))))/(volume/1000)), ((((Samples$CO2_Conc_ppm_dilcorr/1000))/16.04)1000000))

#quick first look at the samples ch4_samples1 <- ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_umol, fill=G_W)) + geom_bar(stat = ‘identity’) + scale_fill_manual(values=c(“darkgrey”,“darkblue”))+ theme_classic() + labs(x= ” “, y=”CH4 (uM)“, title=”CH4: Blue = PW Sample”) + theme(legend.position=“none”)

co2_samples1 <- ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_umol, fill=G_W)) + geom_bar(stat = ‘identity’) + scale_fill_manual(values=c(“darkgrey”,“darkblue”))+ #scale_fill_gradient2(low=‘red’, mid=‘white’, high=‘blue’, space=‘Lab’) + theme_classic() + labs(x= ” “, y=”CO2 (uM)“, title=”CO2: Blue = PW Sample”) + theme(legend.position=“none”)

ggarrange(ch4_samples1, co2_samples1, nrow=1, ncol=2)

```
## Write out processed data & slopes
```

‘‘‘ r

```
#check results
colnames(Samples)
```

```
##  [1] "Machine"         "User"            "Run_Date"        "Sample_Type"
##  [5] "Type1"           "Sample ID"       "Dilution_Factor" "STD_Conc"
```

```
##  [9] "CO2_Area"        "CH4_Area"        "Lab_Notes"       "filename"
## [13] "CH4_Curve"       "CH4_Conc_ppm"    "CO2_Conc_ppm"    "CH4_Flag"
## [17] "CO2_Flag"
```

```
Samples1 <- Samples %>%
  mutate(CH4_Conc_ppm = ifelse(CH4_Conc_ppm < 0, 0, CH4_Conc_ppm)) %>%
  mutate(CO2_Conc_ppm = ifelse(CO2_Conc_ppm < 0, 0, CO2_Conc_ppm))

write.csv(Samples1, "COMPASS_TEMPEST_SGW_2022_Processed.csv")
```

#end